

Expert Opinion

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The cardiovascular risk of tiotropium: is it real?

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Importance of the field: Anticholinergic agents are of noteworthy value in the treatment of chronic obstructive pulmonary disease (COPD), but concerns have been raised about a possible association between their use and cardiovascular (CV) morbidity and mortality. In this review, we have examined whether and why an anticholinergic agent, and in particular tiotropium, might cause CV risks.

Areas covered in this review: We first examine the potential pharmacological mechanisms that justify the CV risk with an anticholinergic agent, and then the main clinical trials, observational (cohort or case-control) studies, descriptive reviews and meta-analyses that have looked at the CV risks associated with long-term tiotropium, which are available in MEDLINE, EMBASE and Cochrane Controlled Trials Register databases, using the following MeSH, full text and keyword terms: tiotropium bromide OR Spiriva AND COPD OR chronic obstructive pulmonary disease.

What the reader will gain: The almost absolute confidence that there is no real increased risk for death or CV morbidity during treatment with this inhaled anticholinergic agent in patients with COPD because of the results of a large 4-year trial and a robust and extensive analysis of > 19,000 patients participating in placebo-controlled tiotropium clinical trials. Nonetheless, because high-risk patients such as those with coronary artery disease, heart failure, cardiac arrhythmia, hypoxemia requiring daytime oxygen therapy and a creatinine > 2 mg/dl were excluded from Phase III clinical trials, it is impossible to exclude these patients from an increased risk of drug-related cardiac events in a real-world setting.

Take home message: Despite the recently raised concerns about an excess risk of CV adverse events with inhaled short-acting anticholinergic agents, the risk:benefit ratio of tiotropium bromide appears still favorable, although it is not known whether high-risk patients are at an increased risk of drug-related CV events.

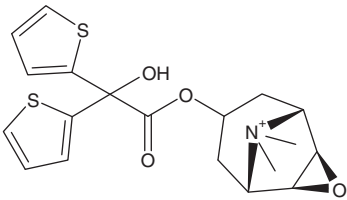
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1. Introduction

Inhaled bronchodilators are the mainstay of the current pharmacological treatment of chronic obstructive pulmonary disease (COPD) [1,2]. COPD guidelines state that regular treatment with long-acting bronchodilators is more effective and convenient than therapy with short-acting bronchodilators [1,2]. In effect, recent large trials, Towards a Revolution in COPD Health [3] and Understanding Potential Long-Term Impacts on Function with Tiotropium (UPLIFT[®]) [4], have documented long-term improvements in lung function, health-related quality of life (HRQL) and possibly survival from the use of long-acting bronchodilators, suggesting an influence on progression of the disease (Box 1) [5].

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Box 1. Drug summary.	
Drug name	Tiotropium
Phase	Launched
Launched indication	Chronic obstructive pulmonary disease
Pharmacology description	Long-acting muscarinic antagonist
Route of administration	Respiratory, inhaled bronchial
Chemical structure	
Pivotal trial(s)	UPLIFT
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However, no distinction is made by current guidelines as to which class of drugs should be considered first [1,2], although we believe that anticholinergic agents are of noteworthy value because of the involvement of parasympathetic cholinergic pathways arising from the vagus nerve in the pathophysiology of airflow obstruction in COPD [6]. Moreover, vagally mediated mechanisms are also implicated in enhanced sub-mucosal gland secretion in patients with COPD [6].

Tiotropium bromide (Spiriva[®], Boehringer Ingelheim, Ridgefield, CT, USA) is the first drug in a new generation of anticholinergic drugs called long-acting antimuscarinic agents because they provide 24-h bronchodilation [7]. It was approved for use in Europe in 2002 and in the US and Canada in 2004. Increases in forced expiratory volume in 1 s (FEV₁) and forced vital capacity values from baseline appear significantly greater with tiotropium bromide than with placebo, the first-generation anticholinergic agent ipratropium bromide or a long-acting β_2 agonist [8]. Available data regarding long-term use of tiotropium for stable COPD suggest a significant reduction of COPD exacerbations and related hospitalizations, as well as improved HRQL [4]. Because of the characteristics of tiotropium bromide, this agent is now considered the first-line drug in the treatment of COPD [8], although there has been recent uncertainty about whether it increases or decreases cardiovascular (CV) risk in the treatment of patients with COPD [9].

In fact, concerns have been raised about possible associations of anticholinergic agents with CV morbidity and mortality. In effect, a body of evidence supports the possible existence of a link between the use of short-acting inhaled anticholinergic agents and CV risk, but it should be noted that this evidence comes from retrospective analyses and as

such there are inherent limitations and problems that preclude definitive conclusions.

In the Lung Health Study, 5887 patients with mild to moderate COPD were randomized to ipratropium bromide (1961), a short-acting anticholinergic agent, or placebo (1962), both with a smoking intervention program, or usual care without a smoking intervention program (1964) over a 5-year period. In the first year of treatment, there was an improvement in FEV₁ with ipratropium relative to placebo [10]. However, over the full 5 years, those assigned to ipratropium had a 3.7-fold increase in the risk of arrhythmias (11 vs 3 episodes, respectively), a 26% increase in the risk of CV events (136 vs 108 events, respectively) and a 2.7-fold increase in fatal CV events (15 vs 6 events, respectively) compared with placebo. Nonetheless, Lanes *et al.* [11] used the Lung Health Study data to compute morbidity and mortality rates according to inhaler use as obtained from subjects at regular visits and observed that the relative risks (RRs) of overall CV morbidity and mortality and of CV mortality specifically were higher in subjects who did not use their inhaler (1.63 (95% CI 1.08 – 2.36) and 3.32 (95% CI 0.91 – 12.07), respectively) than in those who used more than four puffs/day (1.10 (95% CI 0.72 – 1.68) and 1.62 (95% CI 0.39 – 6.79), respectively). A case-control study, which examined associations between medications and risk for death in veterans with newly diagnosed COPD, found that ipratropium was associated with a 34% increase in the odds of CV death (odds ratio (OR) 1.34 (CI 1.22 – 1.47)) [12], but it needs to be noted that the analysis could not be appropriately adjusted for severity as smoking status and FEV₁ were not available. Therefore, we cannot exclude that the entire analysis is confounded by severity. A recent cohort study of 82,717 US veterans with a new diagnosis of COPD between 1999 and 2002 confirmed an increased risk of CV events (acute coronary syndrome, heart failure or cardiac dysrhythmia) associated with the use of ipratropium bromide within the past 6 months, whereas the risk was not present in subjects who received anticholinergics > 6 months prior [13].

However, using a very large database, Sin and Tu [14] did not find an association between ipratropium bromide and mortality in COPD patients after adjustment for potential confounders.

2. Mechanisms that justify the CV risk with an anticholinergic agent

In order to understand if and why an anticholinergic agent can cause CV risks, the potential link between the parasympathetic system and the CV response must be examined.

2.1 Parasympathetic nervous system and heart

All regions of the mammalian heart are innervated by parasympathetic (vagal) nerves. The parasympathetic regulation of cardiac automaticity is mediated by the activation of muscarinic receptors following release of acetylcholine (ACh)

from vagal nerve endings. Muscarinic receptors reside in both the atria and ventricles but have a greater density in the former [15]. The presence of mRNA for both M_2 and M_3 receptors has been demonstrated in the human heart [16,17], M_2 receptors being much more abundant. The protein levels of the M_2 isoform are comparable between human atrium and ventricle, while the density of M_3 appears ~ 10-fold higher in human ventricle than in atrium [18]. Nonetheless, pharmacological, biochemical, immunohistochemical and molecular biological studies using whole heart tissues or isolated cardiomyocytes indicate that the heart, especially cardiomyocytes, also expresses low levels of non- M_2 muscarinic receptors, including the M_1 subtypes [19]. M_3 receptors have also been identified in the coronary circulation of several animal species [20] and may be colocalized with M_4 receptors on various cardiac structures, but their role in modulating measurable physiological responses in humans remains uncertain [21].

ACh stimulates the muscarinic receptors which, in turn, activate NOS through guanylate cyclase to inhibit the L-type calcium channel. The stimulation of cardiac muscarinic receptors provoked by the vagal activation modulates pacemaker activity, atrioventricular conduction, and directly (in atrium) or indirectly (in ventricles) the force of contraction. In general, cholinergic agonists induce a potent negative chronotropic effect on cardiac automaticity and atrioventricular conduction *in vivo* as well as in isolated heart preparations, but the functional response elicited by muscarinic receptor activation depends on species, age, the anatomic structure investigated and the agonist concentration used [20]. The general finding is that low concentrations of muscarinic agonists cause decreases in heart rate, atrioventricular conduction and force of contraction, while high concentrations might under certain circumstances induce increases in heart rate and force of contraction [20].

The M_2 receptor can indirectly activate the muscarinic-gated potassium channel (I_{KACH}) to slow the sinus rate. Additionally, M_2 receptors on the presynaptic sympathetic nerve terminal will inhibit noradrenaline release. The role of the postjunctional M_2 receptor subtype in the production of bradycardia and a decrease in cardiac output is well established. The latter action is due mainly to a reduction in the rate of contraction of the atria, because the ventricles have only a sparse parasympathetic innervation. In effect, in the atrium, stimulation of muscarinic M_2 receptors causes direct negative inotropic and chronotropic effects; in the ventricle, on the other hand, the negative inotropic effect can be only achieved when the basal level of cAMP is elevated by β -adrenoceptor agonists [22]. Thus, the bradycardic effect of carbachol is abolished in atria from M_2 knockout mice [23].

Cardiac M_3 receptors, which mediate a potassium current involved in mammalian heart rate [24], may affect heart rate and cardiac repolarization, modulate inotropic effects, protect against ischemic injury, regulate cell-cell communication, and have antiarrhythmic and proarrhythmic effects, but play a minor role under physiological conditions [25], and in effect

data obtained using knockout mice suggest limited involvement of M_3 receptors in physiological cardiac function [26]. Nonetheless, the role of M_3 receptors might become prominent under pathological situations. It has been documented that the density of M_2 receptors was significantly reduced, accompanied by decreased I_{KACH} current density in atrial cells from the dogs with congestive heart failure (CHF) relative to the control healthy dogs [27]. In sharp contrast, M_3 receptor density was remarkably increased and, correspondingly, the potassium channel I_{KM3} current density was also increased in CHF dogs. These facts suggest that pathological conditions are another factor influencing the expression and relative contribution of the cardiac M_3 receptors. Apparently, increased M_3 muscarinic receptor activity enhances susceptibility to atrial fibrillation [28]. M_3 receptor activation may have an opposite effect on heart rate compared with M_2 receptor activation.

It must be mentioned that M_3 receptors mediate the ACh-dependent vasodilation in mouse coronary circulation [29]. Whether this finding extends to human tissue remains to be determined, and data from functional studies using the human coronary artery are awaited.

2.2 CV consequences of parasympathetic blockade

Suppressed parasympathetic nervous system function in CV diseases was believed to lead to tachycardia and arrhythmia, thus, contributing to mortality in these CV diseases.

The main effect of atropine on the heart is to alter the rate. The dose often decreases transiently with average clinical doses (0.4 – 0.6 mg). It has been suggested that this effect is due to the block of M_1 receptors on postganglionic parasympathetic neurones that relieves the inhibitory effects of synaptic ACh and increases the release of transmitter [30]. Larger doses of atropine cause progressively increasing tachycardia by blocking vagal effects on M_2 receptors on the sinoatrial nodal pacemaker [31]. Unfortunately, the lack of specific M_2 receptor agonists and antagonists has been an obstacle for detailed investigations of the physiological role of the M_2 receptor signal in ventricular physiology and pathophysiology, but there is experimental evidence that deficiency of M_2 receptors increases susceptibility of ventricular function to chronic adrenergic stress [32]. Indeed, when cardiac vagal tone is acutely blocked with atropine, β -adrenergic-mediated increases in heart rate are greater [33], suggesting a blunting effect of vagal tone on the adrenergic response in the heart. However, in conscious dogs, the heart rate after atropine is higher than after bilateral vagotomy, an effect that is not mediated by the β -adrenergic system [34].

Bauer *et al.* [35] showed that in dogs both atropine and ipratropium, when given in intravenous (i.v.) doses of ≥ 5 pg/kg, produced an increase in heart rate of 10% or more. This study also demonstrated that the effect of i.v. ipratropium on heart rate was not concentration-dependent, and that a 10% solution of ipratropium produced a greater effect on heart rate than atropine in the same concentration, while blood pressure was unchanged.

In humans, the mean maximal heart rate rose significantly for atropine 0.03 mg/kg i.v. (+46.2%) and for ipratropium bromide 0.03 mg/kg i.v. (+57.4%) [36]. The effects obtained with ipratropium bromide on the heart rate lasted nearly twice as long as those obtained with atropine (120 ± 38.4 and 70 ± 30 min, respectively, for the pharmacological half-life). In 13 patients with sinus bradycardia, heart rate and hemodynamics were recorded before and 3, 30, 60 and 120 min following i.v. administration of 0.5 mg ipratropium bromide. Heart rate already started to rise during the injection of ipratropium bromide. It increased by an average of 78% three minutes following the administration as compared to pretreatment control values [37]. Mean heart rate still was markedly increased by an average of 26% at 120 min following the injection of the drug. This effect of ipratropium bromide on heart rate was accompanied by an increase in cardiac index while stroke volume decreased due to a decrease in ejection fraction and diastolic filling time.

Nonetheless, acute inhalation of ipratropium bromide appeared to have no significant effect either on cardiac vagal tone or on heart rate, at least among young asthmatics [38]. This is not surprising. In fact, although ipratropium is structurally similar to atropine and has similar actions if given parenterally, it is a quaternary ammonium compound. Therefore, it has low oral bioavailability, but systemic exposure, superior to oral administration, can be achieved by inhalation although it is poorly absorbed from the lung. This is the reason why ipratropium has few extrapulmonary effects, even with very large inhaled doses.

Although most vascular beds lack significant cholinergic innervations, chronic peripheral muscarinic blockade leads to significant hemodynamic alterations, including sustained elevations in blood pressure, heart rate and blood pressure variability [39], which are known risk factors for CV morbidity.

It is intriguing that in experimental heart failure, muscarinic blockade has a more modest effect on heart rate compared with controls [40]. Muscarinic receptor blockade increases cardiac noradrenaline spillover when heart failure is not present, but a blunting of parasympathetic influence on sympathetic activity is present in heart failure [41].

3. Pharmacological profile of tiotropium

Tiotropium is a synthetic, nonchiral quaternary ammonium compound structurally related to ipratropium. It is a specific, highly potent antimuscarinic, demonstrating very slow dissociation from muscarinic receptors, but dissociation from M_2 receptors is faster than from M_3 or M_1 , which in functional *in vitro* studies, appears as kinetic receptor subtype selectivity of M_3 and M_1 over M_2 [42]. After inhalation, the absolute bioavailability of tiotropium is 19.5% in healthy volunteers. It has a volume of distribution of 32 l/kg and is 72% bound to plasma proteins. Furthermore, tiotropium 18 µg inhaled via HandiHaler® results in a C_{5min} of 14 pg/ml (~ 30 pM) and has a terminal phase half-life of 5 – 6 days at

steady-state [43]. It is important to highlight that tiotropium 30 pM is likely to correspond to receptor occupancy of about 6%; peak concentrations of 9 nM (which are not achievable with inhaled doses) would be needed to achieve 95% receptor occupancy [42]. These pharmacological characteristics suggest that tiotropium is not associated with a significantly increased risk for either a cardiac or vascular serious adverse event, including the most common events in the categories of ischemic heart disease and cardiac failure. A 12-week study found no significant differences in heart rate, conduction abnormalities, rhythm or QTc duration on 24-h ECG monitoring over a 12-week period in patients receiving tiotropium, compared with placebo [44].

4. The CV risk associated with long-term tiotropium

In order to evaluate the potential CV risk associated with long-term tiotropium, we examined the main clinical trials, observational (cohort or case-control) studies, descriptive reviews and meta-analyses that have looked at the CV risk associated with long-term tiotropium and which are present in MEDLINE, EMBASE and Cochrane Controlled Trials Register databases using the following MeSH, full text and keyword terms: tiotropium bromide OR Spiriva AND COPD OR chronic obstructive pulmonary disease.

In 2004, Morganroth *et al.* [45] reported a description of the serial electrocardiograms that were incorporated into multiple placebo-controlled clinical trials including long-term (6 and 12-month) trials with tiotropium 18 µg/day ($n = 2128$) and a 4-week dose-ranging study with tiotropium up to 36 µg/day ($n = 169$). In addition, 24-h electrocardiographic (Holter) monitoring was performed as part of a 6-week, placebo-controlled trial with tiotropium 18 µg/day ($n = 121$). In all of these trials, no significant differences were observed in any of the electrocardiogram or Holter outcome parameters compared to placebo. Specifically, there was no clinically relevant difference in heart rate, atrioventricular conduction or the occurrence of ventricular or supraventricular arrhythmias. These findings fit with what was reported by Covelli *et al.* [44].

In 2006, Kesten *et al.* [46] published a pooled analysis of adverse event data from preapproval and postapproval tiotropium clinical trials (19 randomized, double-blind, placebo-controlled studies that included 4435 tiotropium-treated patients and 3384 placebo treated patients contributing 2159 person-years of exposure to tiotropium and 1662 person-years of exposure to placebo). Among heart rate and rhythm disorders, the RRs of tachycardia (excluding ventricular tachycardia and fibrillation) were 1.68 (95% CI 0.69 – 4.11) for any tachycardia and 1.16 (95% CI 0.33 – 4.03) for serious tachycardia. The RR of other arrhythmias such as bradycardia, irregular heart beat and extrasystoles in patients receiving tiotropium compared with patients receiving placebo was 2.71 (95% CI 1.10 – 6.65). There

was no apparent excess of other arrhythmias classified as serious (RR 0.92; 95% CI 0.27 – 3.14). The RR of left-heart failure among patients receiving tiotropium compared with patients receiving placebo was 0.46 (95% CI 0.21 – 1.00). There was no increase in total and serious events due to ischemic heart disease.

A further publication [47] demonstrated that, in patients with COPD, inhalation of tiotropium reduced heart rate by 7 bpm at rest and throughout exercise, and this was accompanied by an increase in cardiac stroke volume with a reduction of the rate-pressure product (myocardial work). The reduction in heart rate correlated best with reductions in lung hyperinflation, suggesting a cardio-pulmonary link. Given the very low peripheral concentrations of tiotropium following inhalation, this suggests that the likely explanation for this effect is not a direct CV effect but an effect on cardio-pulmonary interactions.

Using the Danish healthcare registries, de Luise *et al.* [48] documented that tiotropium was associated with reduced total and respiratory mortality, and was not associated with increased total hospitalizations, or hospitalizations from cardiac causes. Adjusted RR estimates for cardiac hospitalizations were imprecise but elevated for atrial fibrillation and atrial flutter (RR = 1.21, 95% CI: 0.87, 1.69), but not for angina (RR = 1.01, 95% CI: 0.68, 1.60), myocardial infarction (RR = 1.05, 95% CI: 0.69, 1.60) or CHF (RR = 1.03, 95% CI: 0.73, 1.47). There were few cases of supraventricular tachycardia (RR = 0.65, 95% CI: 0.26, 1.65) and ventricular arrhythmia (RR = 1.97, 95% CI: 0.56, 6.88), and results for these end points were imprecise.

All these data were reassuring on the CV safety of tiotropium in COPD patients. However, a systematic review with meta-analysis of 17 clinical trials (12 trials evaluated inhaled tiotropium vs control therapy and 5 trials evaluated inhaled ipratropium vs control therapy) reported an increased risk of myocardial infarction (RR 1.53, 95% CI 1.05 – 2.23), CV death (RR 1.80, 95% CI 1.17 – 2.77) and a trend towards increased risk of stroke (RR 1.46, 95% CI 0.81 – 2.62) [49]. All-cause mortality was reported for 2% of patients treated with inhaled anticholinergics compared to control patients (1.6%) (RR 1.26, 95% CI 0.99 – 1.61). In addition, a sensitivity analysis of five long-term trials (6 months) confirmed that inhaled anticholinergics significantly increased the risk of CV death, myocardial infarction or stroke (RR 1.73, 95% CI 1.27 – 2.36). Further, the results of this meta-analysis showed that the use of inhaled anticholinergics for > 30 days increased the risk of CV death, myocardial infarction or stroke in patients with COPD, by 58%. This conclusion seemed to confirm the meta-analysis of Barr *et al.* [50], in which the frequency of arrhythmia was significantly higher with tiotropium than with placebo when adjusted for statistical heterogeneity (OR 2.33; 95% CI 1.11 – 4.88) and also the warning issued by the FDA about a potential increased risk of stroke in patients using tiotropium [51].

Moreover, a 2-year multi-center, randomized, double-blind, double-dummy trial that compared salmeterol/fluticasone 50/500 mg twice daily and tiotropium 18 mg/day documented that fatal cardiac events occurred in 1% of the salmeterol/fluticasone group compared to 3% in the tiotropium group [52].

Fortunately, the large UPLIFT trial, in which tiotropium was compared to placebo in 5993 patients followed for 4 years [4], has clarified the doubts generated by these meta-analyses. In this study, CV adverse events were significantly less with tiotropium (RR 0.84; 95% CI 0.73 – 0.98), although 74% of patients reported having received inhaled corticosteroids, 72% long-acting β -agonists and 46% a fixed combination of the two [4]. The RR of CHF (RR 0.59; 95% CI 0.37 – 0.96) and myocardial infarction (RR 0.71; 95% CI 0.52 – 0.99) was also significantly lower with tiotropium compared to placebo, although those of angina (RR 1.44; 95% CI 0.91 – 2.24) and cardiac failure (RR 1.25; 95% CI 0.84 – 1.87) were not significantly higher with tiotropium compared to placebo. The RR of stroke in the tiotropium versus placebo groups was not significantly different (RR 0.95; 95% CI 0.70 – 1.29). It is important to highlight that the mortality adjudication committee, consisting of three independent physicians (two pulmonologists and one cardiologist) who were not UPLIFT investigators, only recorded myocardial infarction where documented objective evidence was provided in the available record [53]. This decreased the number of cardiac events noted by the investigator. For example, in the on-treatment analysis, there were 36 investigator-reported myocardial infarction events, compared with 17 after adjudications by the committee.

Furthermore, Grosso *et al.* [54] studied the association between inhaled tiotropium bromide and stroke applying the self-controlled case-series method, which helps reduce confounding (by eliminating fixed-effect confounders such as comorbidities) in observational studies investigating drug exposure and adverse outcomes, on routinely collected information from the large UK General Practice Research Database and found no evidence of an increased risk of all-cause stroke for individuals exposed to tiotropium. The calculated age-adjusted incidence rate ratio stroke with tiotropium was highest at 1.1 (compared with fluticasone propionate/salmeterol xinafoate at 1.0 and ipratropium bromide at 0.8), albeit with overlapping CIs. The authors also re-analyzed the data set including only trials involving tiotropium and reporting stroke, and included the UPLIFT trial data [4]. Their analysis also failed to demonstrate a statistically significant effect for tiotropium alone on stroke outcome.

Also, Oba *et al.* [55] reanalyzed the meta-analysis conducted by Singh *et al.* [49], incorporating the results of UPLIFT study [4]. In particular, they further investigated the possible association between tiotropium and increased risk of CV events and documented that the association between inhaled anticholinergics (ipratropium and tiotropium) and an increased risk for a composite of myocardial infarction,

stroke and CV death was not significant with the use of tiotropium.

Salpeter [56] reviewed the available evidence and confirmed that, although the meta-analysis of randomized trials observed that a higher proportion of participants receiving inhaled anticholinergic agents had major CV events compared with those receiving placebo, this differential is eliminated when the large UPLIFT trial [4] is included in the meta-analysis or when evaluating the rate of events per total person-years of exposure. In fact, updated pooled trial data indicate that tiotropium bromide may actually reduce the incidence rates of major CV events and total mortality over time.

An independent systematic review according to the QUOROM statements [57], which evaluated the safety of regular use of inhaled tiotropium bromide in patients with any severity of COPD, confirmed that, compared with control (placebo or salmeterol), tiotropium does not significantly increase the risk of adverse major CV events among COPD patients, although a subgroup analysis suggested that smoking history can modify the risk of CV adverse events [58].

A further decisive and conclusive reassurance on the CV safety of tiotropium comes from the recent robust and extensive analysis published by Celli *et al.* [59]. These authors revised all 30 trials that had the following criteria: ≥ 4 weeks, randomized, double-blind, parallel-group and placebo-controlled. There were 19,545 patients randomized with 8699 receiving placebo and 10,846 receiving tiotropium. Cumulative exposure to study drug (patient-years) was 13,146 patient-years for tiotropium and 11,095 for placebo. The RR (95% CI) for a fatal cardiac event (tiotropium/placebo) was 0.77 (0.58 – 1.03) and for a fatal vascular event it was 0.44 (0.19 – 1.02). Tiotropium showed a trend toward a reduced risk for a cardiac or vascular fatal adverse event. For myocardial infarction and stroke, the RRs (95% CI) were 0.74 (0.41 – 1.34) and 1.02 (0.49 – 2.12), respectively. There was a 17% reduction in the risk of a major CV event in the tiotropium group relative to the placebo group and a 23% reduction for fatal CV events, with both values being statistically significant.

Nonetheless, it should be mentioned that three 48 week clinical trials with tiotropium delivered by the Respimat[®] inhaler in patients with COPD suggested an increase in mortality or, at least, a mortality imbalance in favor of placebo, but with no obvious other safety signal that could be linked to death [60]. It seems to be intriguing that Respimat allows larger tiotropium plasma concentrations than HandiHaler [60]. Obviously, safety data from the Respimat trials are impatiently awaited to better understand this issue.

5. Expert opinion

The analysis of the literature raises serious questions for clinicians who must treat patients with COPD. Bronchodilators are central to the symptomatic treatment of COPD [1,2], but some evidence indicates and both β_2 -agonists and

anticholinergic agents increase the risk for adverse CV events [12,49,61-63]. A cohort study using the UK Health Information Network primary care database documented that the 1061 tiotropium users and 1801 long-acting β -agonists users were similar with regard to the risk of most cardiac events, including angina (HR 0.77; 95% CI 0.37, 1.59), atrial fibrillation or flutter (HR 0.60; 95% CI 0.25, 1.42), myocardial infarction (HR 1.29; 95% CI 0.45, 3.66) and tachycardia (HR 0.66; 95% CI 0.29, 1.51) [64]. Though imprecise, there was evidence of a decreased risk of heart failure (HR 0.65; 95% CI 0.37, 1.12) in tiotropium users.

The fact that both β -agonists and anticholinergics increase CV risk in patients with COPD deserves some comments.

First of all, we cannot exclude that COPD amplifies the impact of bronchodilators on the heart and is a confounding factor when the impact of these agents on the heart is evaluated. In COPD patients, the activity of cardiac sympathetic nerves may be affected by recurrent hypoxemia, hypercapnia, change in airway tone, increased intrathoracic pressure due to airway obstruction, and highly fluctuating heart rate and blood pressure due to increased respiratory effort [65]. The existence of left ventricle myocardial sympathetic nervous alterations due to generalized sympathetic autonomic nervous system overactivity has been suggested [66]. It is logical to assume that administration of β_2 -agonists or anticholinergic bronchodilators may further increase the sympathetic nervous activity of COPD patients and thereby amplify their risk for poor CV events. In any case, patients with hypoxic COPD have also evidence of a subclinical parasympathetic autonomic neuropathy [67].

We cannot, then, forget that a body of evidence indicates that patients suffering from COPD may be at an increased risk of CV complications probably because of shared CV risk factors and underlying systemic inflammation along with the significant numbers of these individuals who continued to smoke during the observation period [67]. Using healthcare databases maintained by the government of Saskatchewan, Canada, Curkendall *et al.* [68] found that the prevalence of all CV diseases was higher in the COPD group than in the comparison group. After adjusting for CV risks, odd ratios of prevalence were: arrhythmia 1.76, angina 1.61, acute myocardial infarction 1.61, CHF 3.84, stroke 1.11 and pulmonary embolism 5.46. Although it must be mentioned that bronchodilators apparently impose a larger RR in people with unrecognized CV disease than in people with known disorders [69], is likely that the increased risk for CV morbidity and mortality is a consequence of COPD and the effect attributed to bronchodilators is just an epiphenomenon. In partial support of this, there is the documentation that, at least in experimental heart failure, muscarinic blockade has a more modest effect on heart rate compared with controls [41].

We fully agree with the opinion of Rabe [70] who felt that, because these drugs are poorly absorbed by the gastrointestinal tract and the lungs, systemic adverse effects such as tachycardia are considered unlikely [6], and prescription habits might have

been biased toward administering this class of drugs to patients with a higher perceived CV risk. In any case, we must also highlight that although several papers have highlighted potential concerns between anticholinergics and CV safety in patients with COPD, there have been conflicting data concerning the CV risk associated with these agents, and it is noteworthy that none of the examined studies had CV outcomes as the primary outcome and the data reported are based on adverse event reporting, not efficacy end points. This means that CV outcomes may not have been prospectively defined in a uniform fashion across the trials and, consequently, we believe, that there may have been important discrepancies between the trials in reporting these outcomes. Also of note is the fact that patients with a recent history of myocardial infarction, arrhythmia or heart failure were excluded from UPLIFT [4] (and possibly other studies); therefore, it is not known whether these patients may be at an increased risk of any drug-related CV events in a real-world setting.

Our opinion is that the literature does not document a real increased risk for death or CV morbidity during treatment with inhaled anticholinergic agents in patients with COPD. In particular, despite the recently raised concerns about an excess risk of CV adverse events with inhaled short-acting anticholinergic agents, the risk:benefit ratio of tiotropium bromide appears still favorable given the favorable safety profile demonstrated in the UPLIFT study [4]. We recognize that the meta-analysis published by Singh *et al.* [49] suggested an association between inhaled anticholinergics and an increased risk of CV death and myocardial infarction and stroke. However, we are of the same opinion of Heredia [71] when he affirms that the methodology of the study of Singh *et al.* [49] raises several important concerns. First, placebo-controlled trials were considered together with active controlled trials, implicitly assuming that the comparator drug is interchangeable with a placebo. Second, the analysis does not take into account the fact that in most trials more patients in the placebo group dropped out than patients taking active medication and that they were followed for briefer periods of time, during which adverse events were reported. Third, most of the evidence in the analysis is from the Lung Health Study [11], in which most of the CV deaths occurred among patients who were not using their inhalers; clearly, these deaths could not be due to a medication they were not taking. Finally, the data set analyzed appears to have an important methodological flaw that potentially invalidates the findings: it appears to include two studies twice, thus, double-counting more than a thousand patients. Also, the FDA Pulmonary-Allergy Drugs Advisory Committee [60] agrees that due to the potential methodological limitations of the meta-analysis published by Singh *et al.* [49], UPLIFT study [4] may provide stronger evidence on the safety of tiotropium.

At this point, we must highlight that almost all information on the potential CV risk caused by anticholinergic comes from meta-analyses. Peer-reviewed meta-analyses of aggregate published data have their appropriate place in scientific

research. However, the large majority of published meta-analyses have been based on summary data from trial publications and this is a critical inadequacy. Moreover, as correctly stressed by Celli *et al.* [59], there are other important limitations to pooling of clinical trial data. There are differences in populations, study design, duration of trials, collection of data and the ability to adjust for differences in exposure. Furthermore, different study parameters exist in different trials, and factors such as illness severity, comorbid conditions and concomitant treatments can have effects on outcome additional to the intervention being studied. In addition, it is recognized that the data are dependent on the physician diagnostic reporting and this may be difficult when elderly patients with COPD become acutely ill. Altogether, literature-based meta-analysis can lead to seriously biased assessments and should be avoided. What is surprising is the fact that nobody has calculated the number needed to treat (NNT), which can be useful to represent the magnitude of the treatment harm in clinically relevant terms. The NNT is calculated as the inverse of the absolute risk difference between treatment groups and provides an estimate of how many people need to receive a treatment before one additional person would experience the harmful outcome.

For all these reasons, we still consider the use of anticholinergic agents a first choice in patients with COPD and concomitant CV disease [72], although we must honestly agree with Oba *et al.* [55] that one limitation in inhaled anticholinergic bronchodilator CV safety database is the paucity of information from high-risk patients such as those with coronary artery disease, heart failure, cardiac arrhythmia, hypoxemia requiring daytime oxygen therapy and creatinine > 2 mg/dl because such patients were excluded from Phase III clinical trials. These patients may be at increased risk of drug-related cardiac events in a real-world setting. Therefore, only an adequately designed randomized trial prespecified on CV outcomes will allay the CV safety concerns with inhaled anticholinergics.

Moreover, we must highlight that there is an increasing body of evidence that is documenting that anticholinergic agents (at least atropine) can decrease heart rate turbulence (HRT), a short-term change in heart rate after ventricular premature beat, presumably caused by the baroreceptor mechanism. It has been shown to be a powerful predictor of mortality in patients with acute myocardial infarction and CHF. Some evidence indicates that normal vagal influence on the sinus node is necessary for normal HRT, and suggests that abnormal HRT may be caused by pathologic loss of vagal activity [73]. Although a direct link between abnormal cardiac vagal influence and increased mortality has not been proven yet, further investigation into the relations among cardiac autonomic activity, HRT, anticholinergic agents and mortality is warranted [74].

In any case, even if it is true that more data need to be available, clinicians should always carefully weigh safety and efficacy data and help patients make informed decisions about their COPD care. All those who should treat patients with COPD need to have an answer to a simple but

fundamental question: if we must consider withholding anticholinergic bronchodilators from patients with mild to moderate symptoms of COPD who are at high risk of a CV event, how should we treat these patients in view of the fact that in patients with CV diseases, the prescription of an inhaled β -agonist must always be undertaken with caution [60] as these agents may precipitate the concomitant cardiac disease [75]?

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Declaration of interest

M Cazzola has received fees for speaking and consulting and/or financial support for attending meetings from Boehringer Ingelheim, and Pfizer.

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